PAPER ID-310858

Roll No:

# BTECH
## (SEM V) THEORY EXAMINATION 2025-26
## MACHINE LEARNING TECHNIQUES

**TIME: 3 HRS** M.MARKS: 70

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

### SECTION A

**1. Attempt *all* questions in brief.** 02 x 7 = 14

| Q no. | Question | CO | Level |
|---|---|---|---|
| a. | Define Machine Learning. | CO1 | K1 |
| b. | What is a well-defined learning problem? | CO1 | K2 |
| c. | State any two applications of Logistic Regression. | CO2 | K2 |
| d. | What is Entropy in Decision Tree Learning? | CO3 | K2 |
| e. | Define Support Vector Machine (SVM). | CO2 | K1 |
| f. | What is backpropagation in neural networks? | CO4 | K2 |
| g. | What is Q-learning? | CO5 | K2 |

### SECTION B

**2. Attempt any *three* of the following:** 07 x 3 = 21

| | | | |
|---|---|---|---|
| a. | Explain types of learning in Machine Learning with suitable examples. | CO1 | K2 |
| b. | Describe Linear Regression and its assumptions. | CO2 | K3 |
| c. | Explain Naïve Bayes classifier with an example. | CO2 | K3 |
| d. | Discuss the properties and issues of Support Vector Machines. | CO3 | K3 |
| e. | Explain k-Nearest Neighbour (k-NN) algorithm and its limitations. | CO3 | K3 |

### SECTION C

**3. Attempt any *one* part of the following:** 07 x 1 = 07

| | | | |
|---|---|---|---|
| a. | Explain the ID3 algorithm for Decision Tree learning. | CO3 | K3 |
| b. | Discuss inductive bias and inductive inference in decision trees. | CO3 | K4 |

**4. Attempt any *one* part of the following:** 07 x 1 = 07

| | | | |
|---|---|---|---|
| a. | Explain perceptron and multilayer perceptron with neat diagram. | CO4 | K3 |
| b. | Derive the Backpropagation algorithm for training neural networks. | CO4 | K4 |

**5. Attempt any *one* part of the following:** 07 x 1 = 07

| | | | |
|---|---|---|---|
| a. | Explain the architecture of Convolutional Neural Networks (CNN). | CO4 | K4 |
| b. | Discuss applications of Deep Learning with suitable case studies. | CO3 | K5 |

**6. Attempt any *one* part of the following:** 07 x 1 = 07

| | | | |
|---|---|---|---|
| a. | Explain Reinforcement Learning with Markov Decision Process. | CO5 | K4 |
| b. | Describe Q-Learning algorithm and its applications. | CO5 | K4 |

**7. Attempt any *one* part of the following:** 07 x 1 = 07

| | | | |
|---|---|---|---|
| a. | Explain Genetic Algorithm cycle with suitable example. | CO5 | K4 |

https://www.pyqonline.com